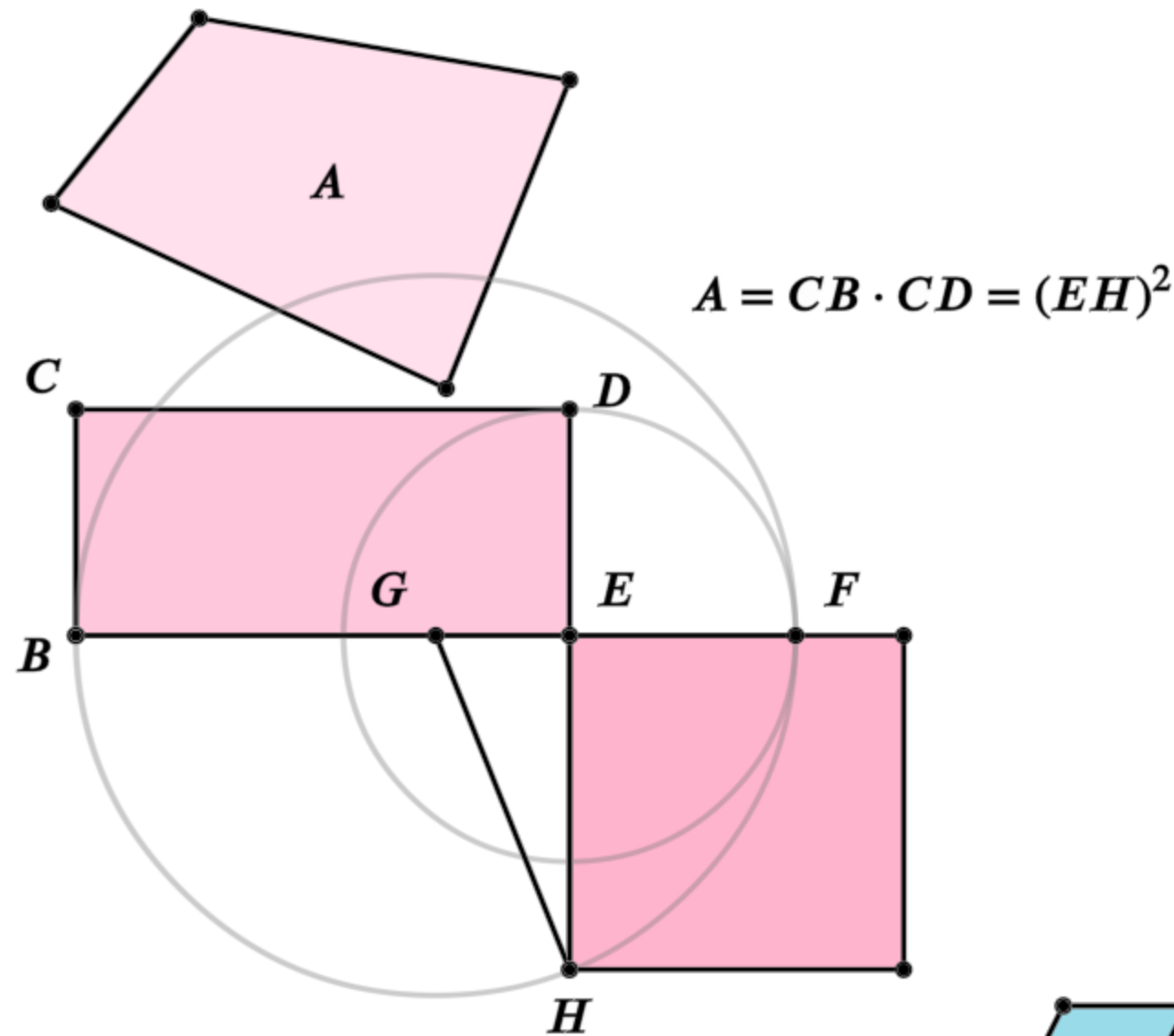


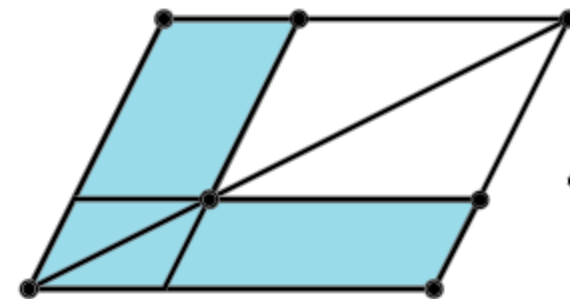
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



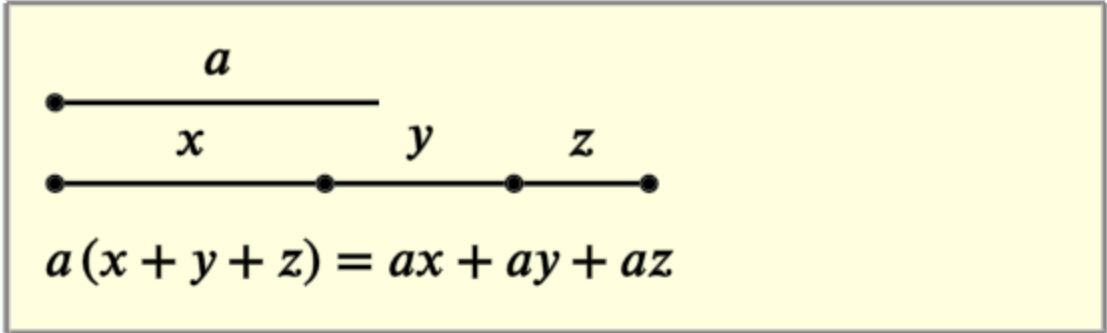
Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

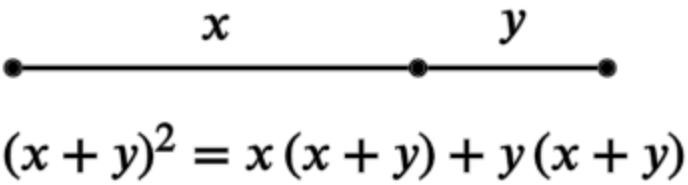
And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

Table of Contents - Book 2

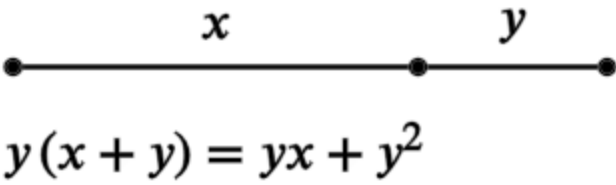
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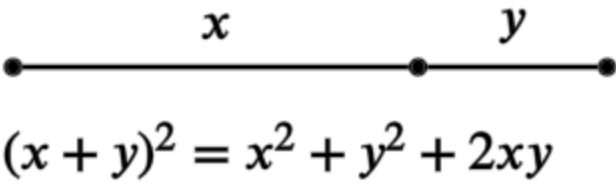
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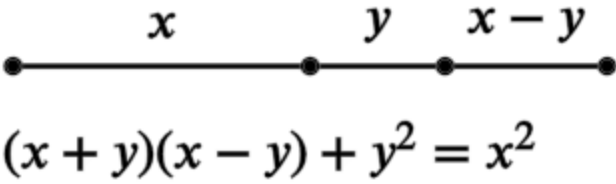
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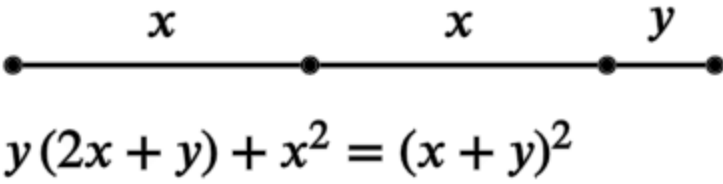
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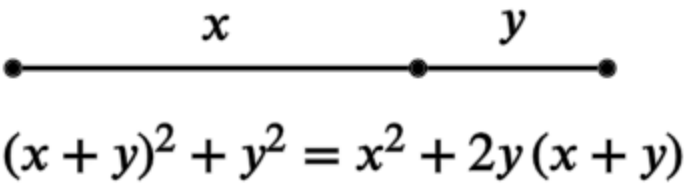
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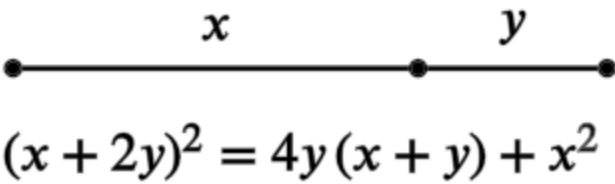
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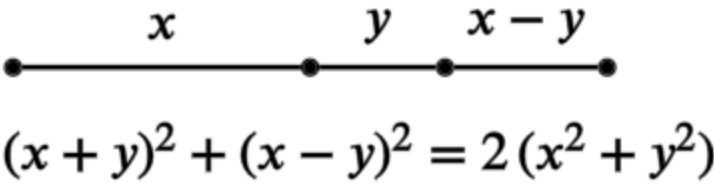
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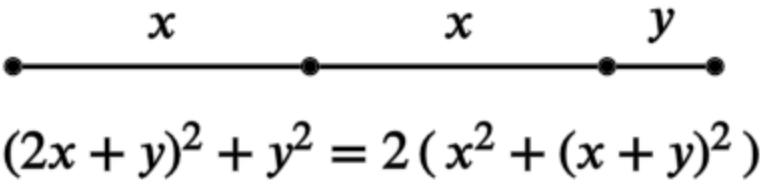
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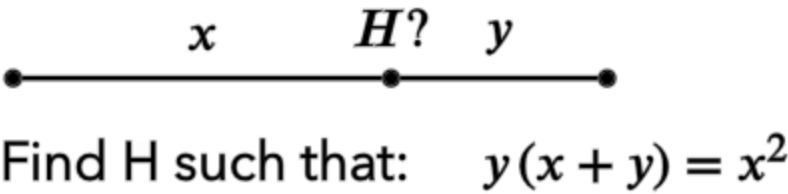
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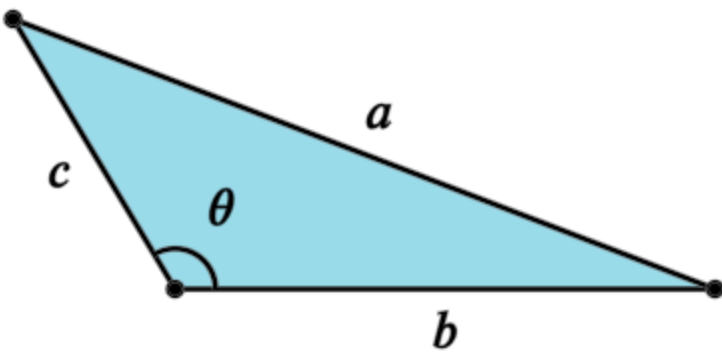
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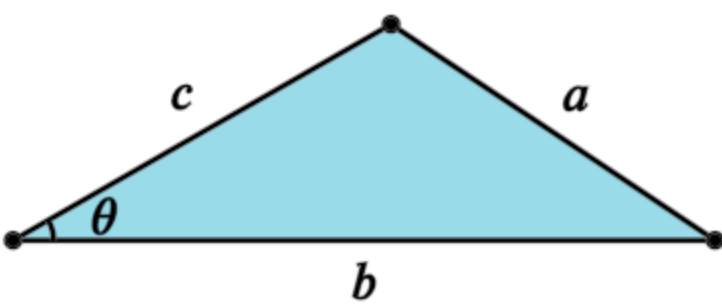
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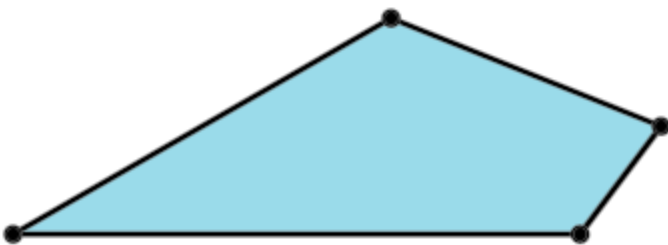
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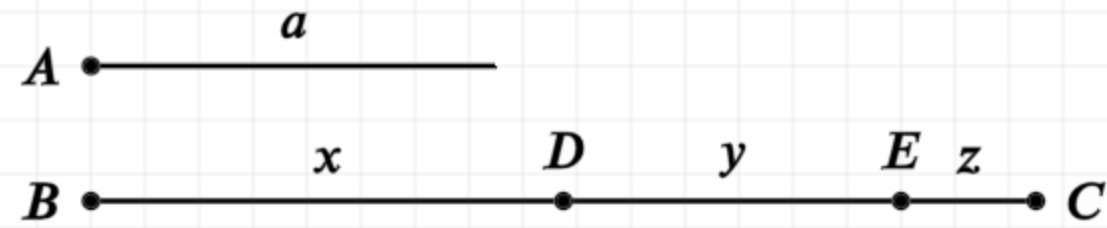


14.



Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



In other words

Let A and BC be two straight lines

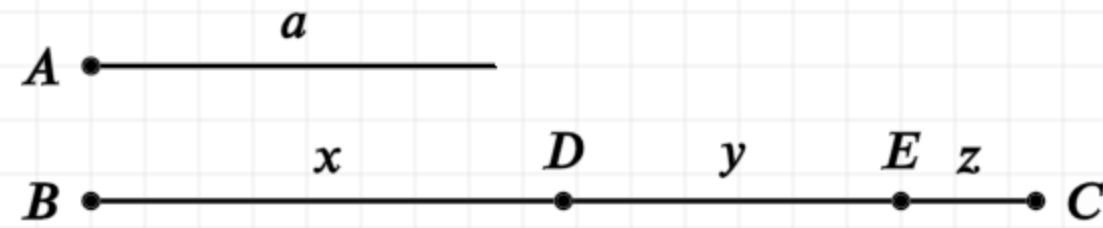
Let BC be arbitrarily cut at points D and E

$$BC = BD + DE + CE$$

$$b = x + y + z$$

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

Then the area of the rectangle formed by line A and BC is equal in area to the sum of the rectangles formed by line A and BD, line A and DE, and line A and EC

$$BC = BD + DE + CE$$

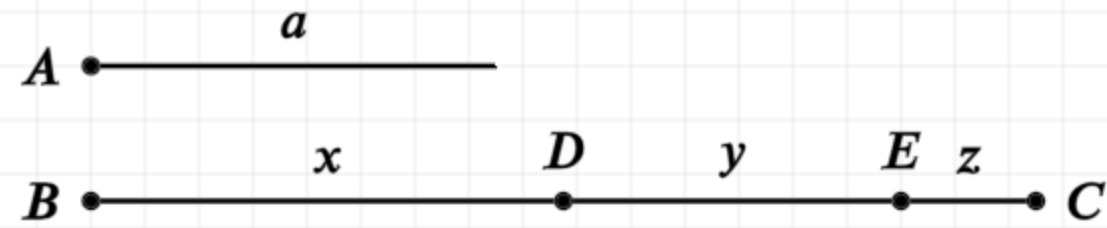
$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

$$b = x + y + z$$

$$ab = ax + ay + az$$

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

$$BC = BD + DE + CE$$

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.

In other words

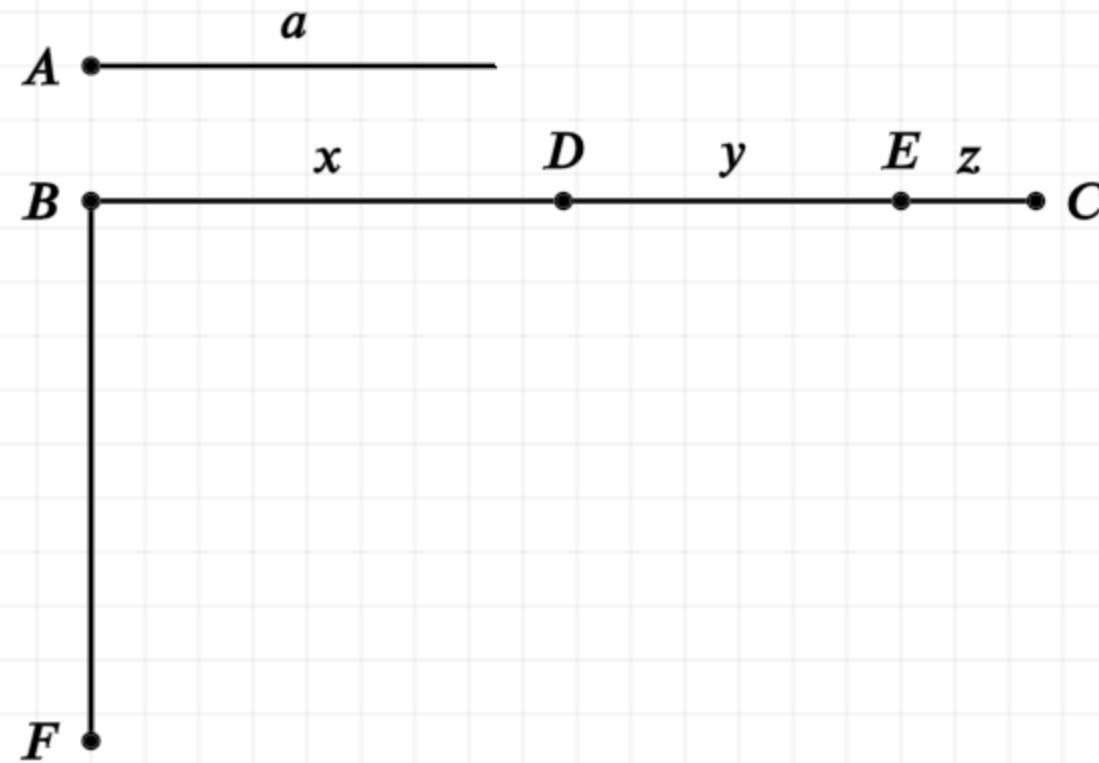
Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)



$$BC = BD + DE + CE$$

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.

In other words

Let A and BC be two straight lines

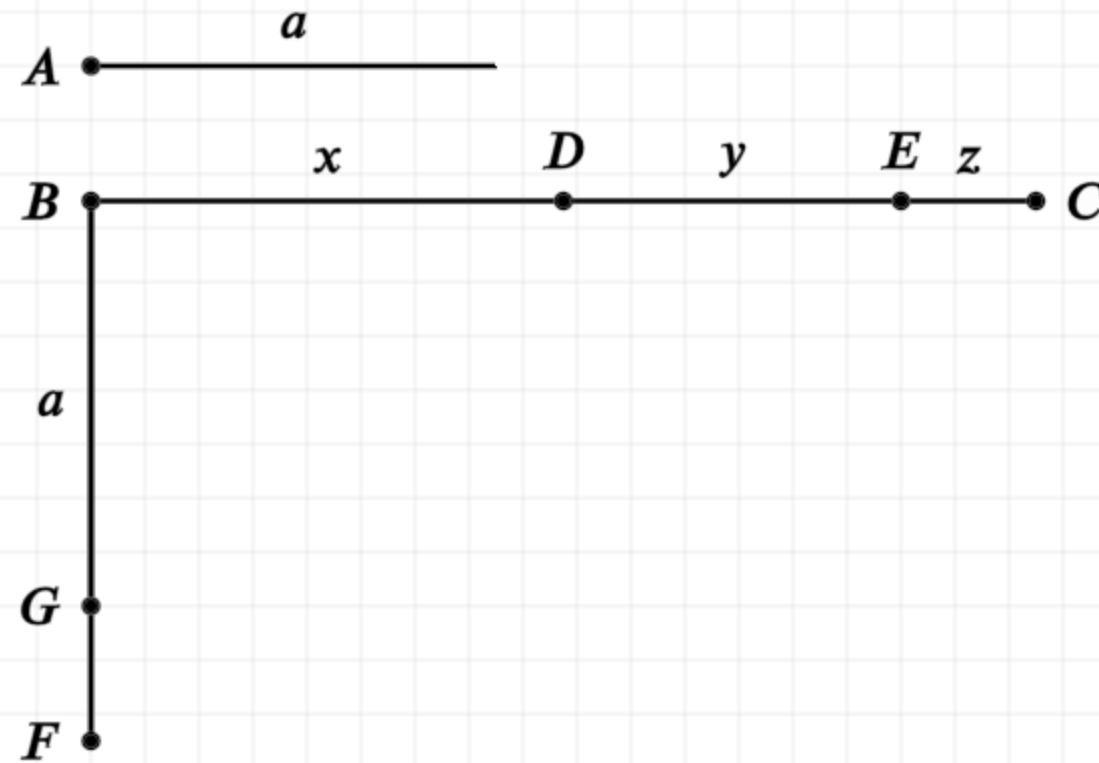
Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

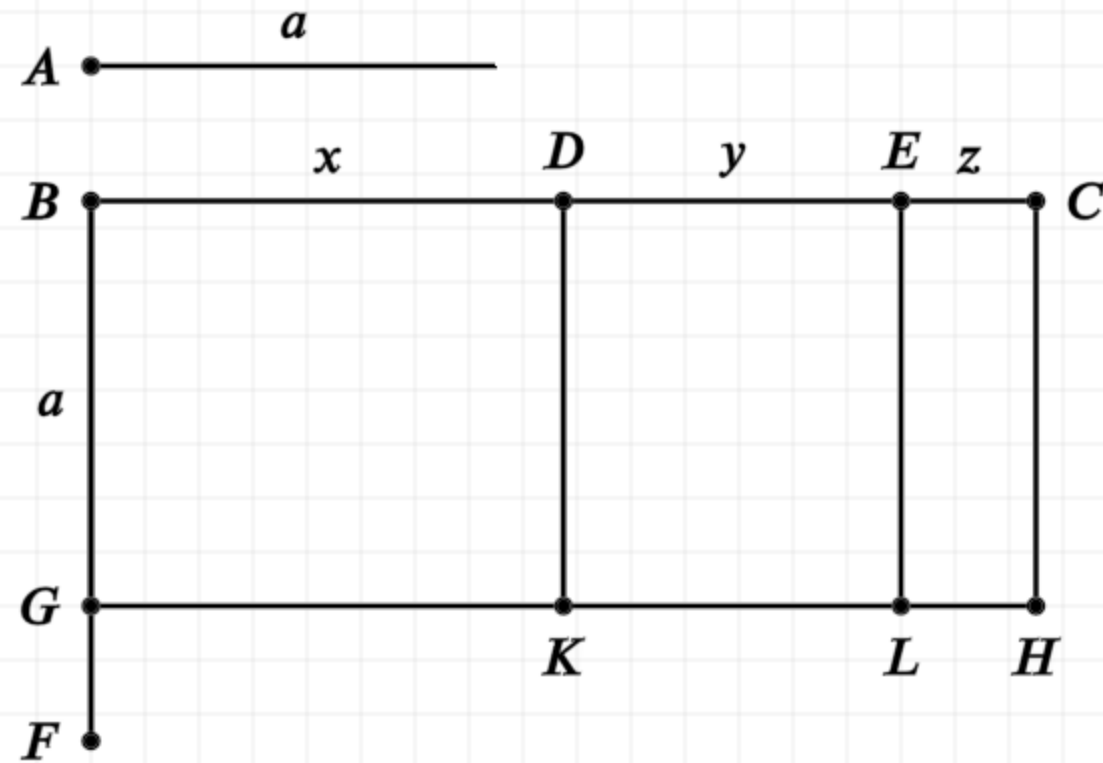


$$BC = BD + DE + CE$$

$$A = BG$$

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

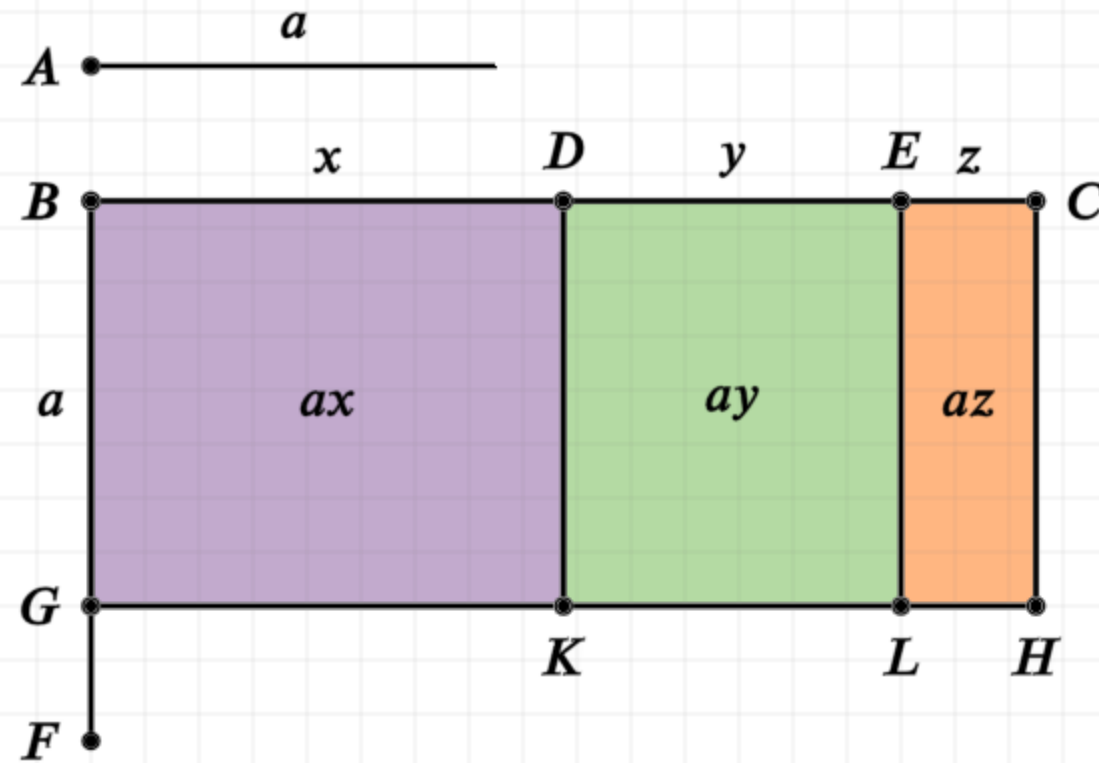
Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

$$\text{■ } BH = \text{■ } BK + \text{■ } DL + \text{■ } EH$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)

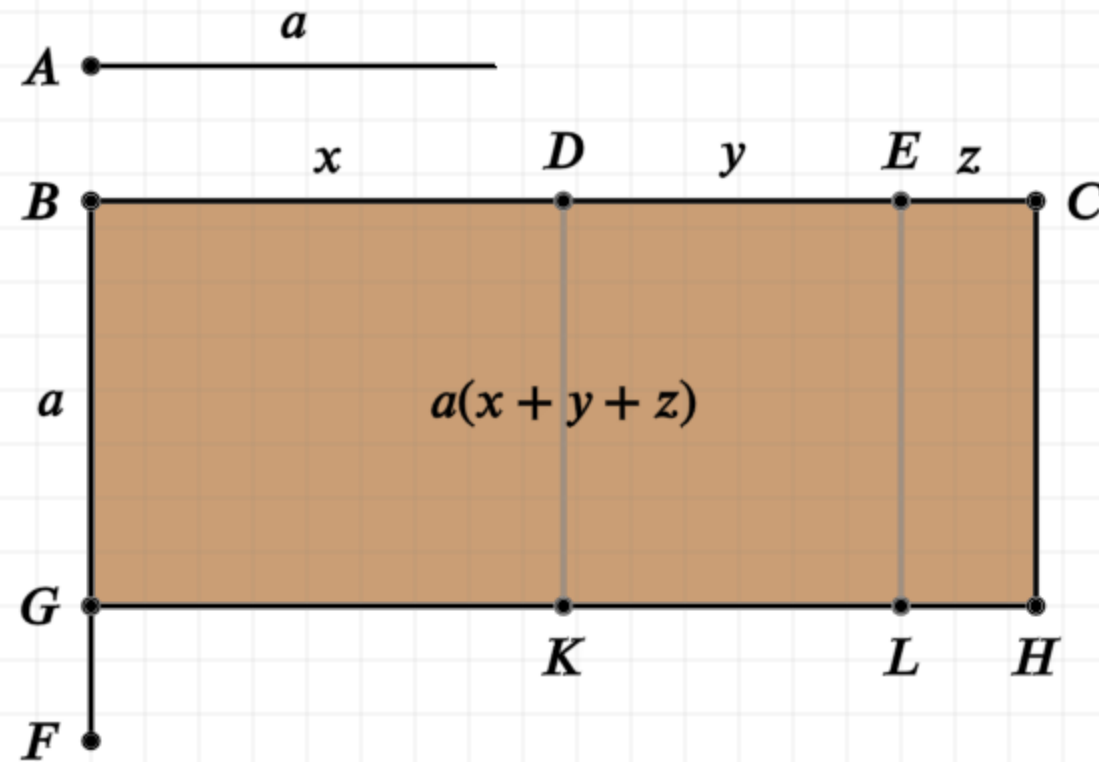
Define point G such that BG equals A (I.3)

Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

The rectangle BH is the sum of the rectangles BK DL and EH

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

$$BH = BK + DL + EH$$

$$BH = BG \cdot BC, \quad \therefore BH = A \cdot BC$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

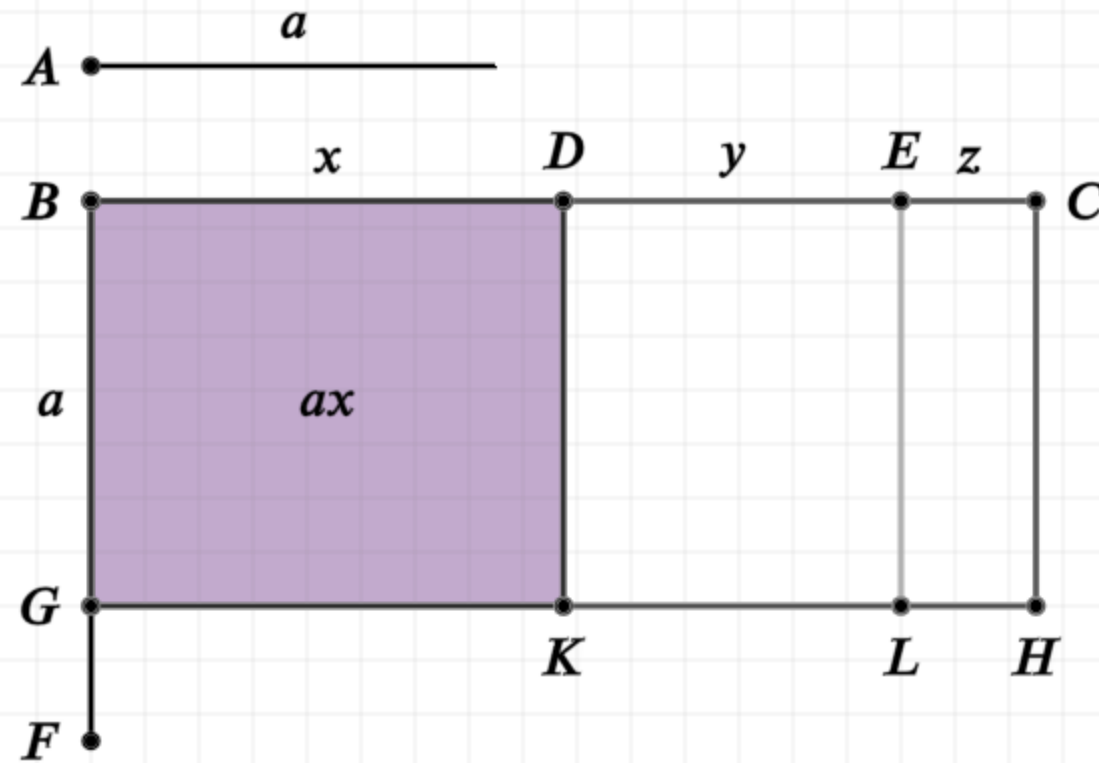
Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

The rectangle BH is the sum of the rectangles BK DL and EH

Since BG is equal in length to A, the rectangle BH is equal to the rectangle contained by lines A and BC

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

$$\text{rectangle } BH = \text{rectangle } BK + \text{rectangle } DL + \text{rectangle } EH$$

$$\text{rectangle } BH = BG \cdot BC, \quad \therefore \text{rectangle } BH = A \cdot BC$$

$$\text{rectangle } BK = BG \cdot BD, \quad \therefore \text{rectangle } BK = A \cdot BD$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

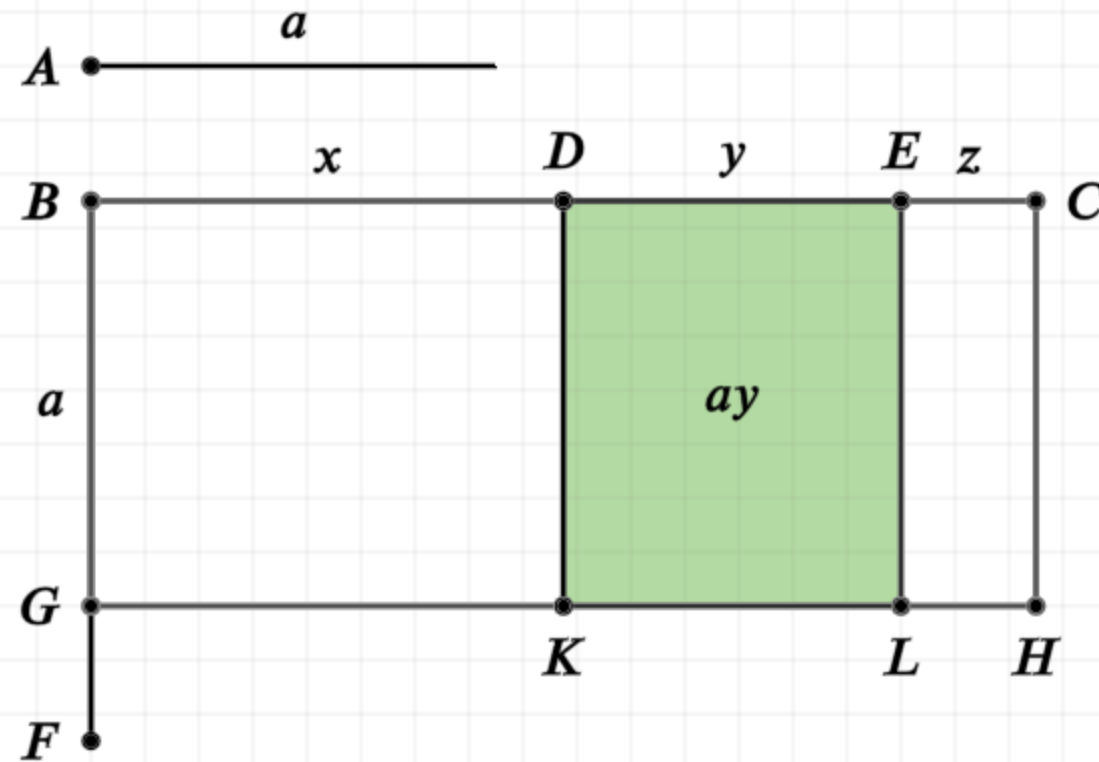
The rectangle BH is the sum of the rectangles BK DL and EH

Since BG is equal in length to A, the rectangle BH is equal to the rectangle contained by lines A and BC

Similarly, the rectangle BK is equal to the rectangle contained by lines A and BD

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

$$\text{Rectangle } BH = \text{Rectangle } BK + \text{Rectangle } DL + \text{Rectangle } EH$$

$$\text{Rectangle } BH = BG \cdot BC, \quad \therefore \text{Rectangle } BH = A \cdot BC$$

$$\text{Rectangle } BK = BG \cdot BD, \quad \therefore \text{Rectangle } BK = A \cdot BD$$

$$\text{Rectangle } DL = DK \cdot DE, \quad \therefore \text{Rectangle } DL = A \cdot DE$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

The rectangle BH is the sum of the rectangles BK DL and EH

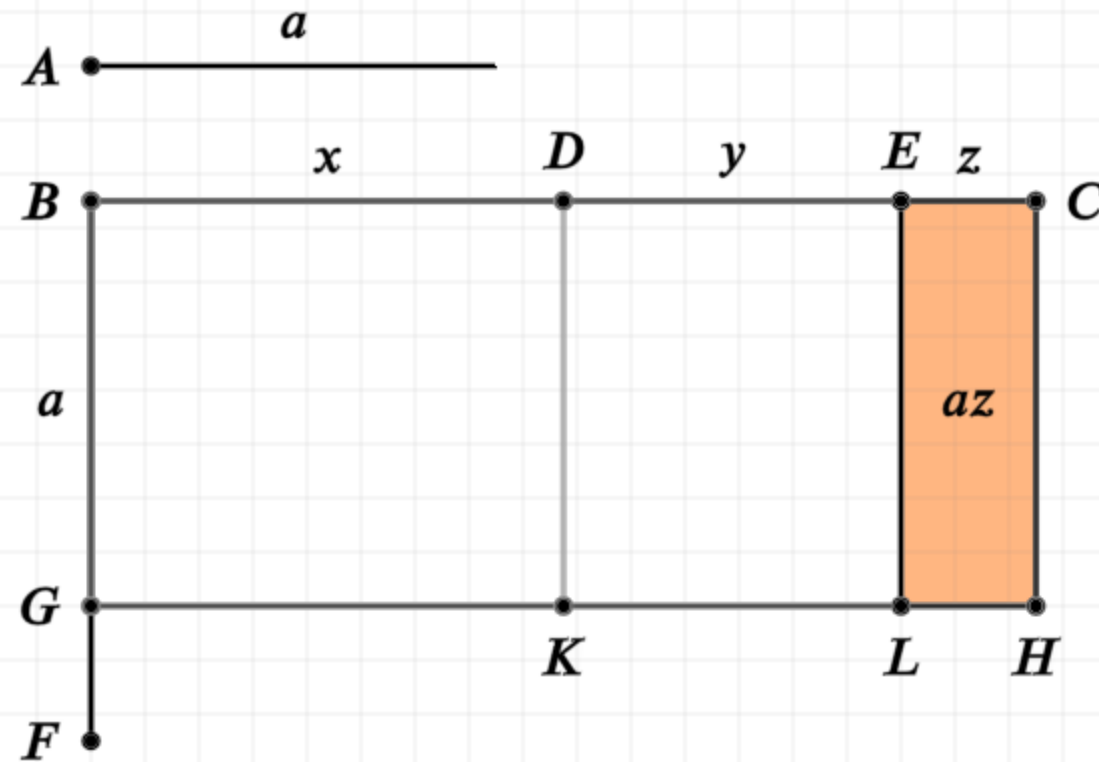
Since BG is equal in length to A, the rectangle BH is equal to the rectangle contained by lines A and BC

Similarly, the rectangle BK is equal to the rectangle contained by lines A and BD

Since BG equals DK (I.34), DL is equal to the rectangle contained by lines A and DE

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

$$\text{Rectangle } BH = \text{Rectangle } BK + \text{Rectangle } DL + \text{Rectangle } EH$$

$$\text{Rectangle } BH = BG \cdot BC, \quad \therefore \text{Rectangle } BH = A \cdot BC$$

$$\text{Rectangle } BK = BG \cdot BD, \quad \therefore \text{Rectangle } BK = A \cdot BD$$

$$\text{Rectangle } DL = DK \cdot DE, \quad \therefore \text{Rectangle } DL = A \cdot DE$$

$$\text{Rectangle } EH = EL \cdot EC, \quad \therefore \text{Rectangle } EH = A \cdot EC$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

The rectangle BH is the sum of the rectangles BK DL and EH

Since BG is equal in length to A, the rectangle BH is equal to the rectangle contained by lines A and BC

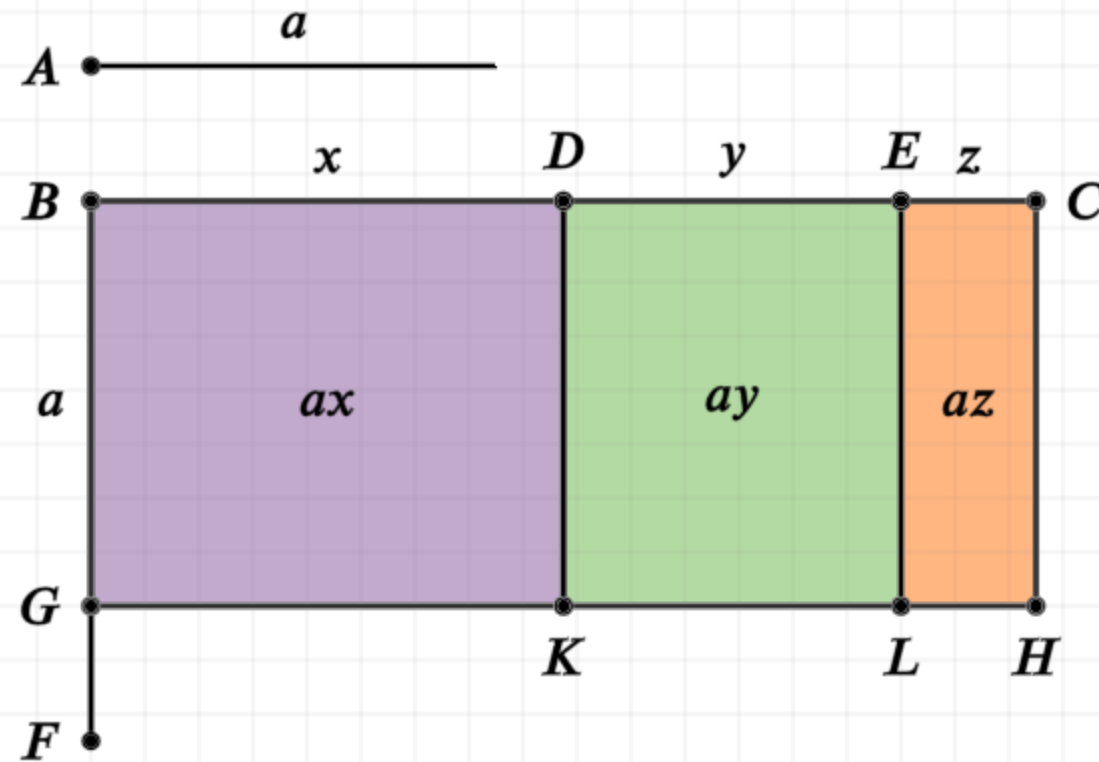
Similarly, the rectangle BK is equal to the rectangle contained by lines A and BD

Since BG equals DK (I.34), DL is equal to the rectangle contained by lines A and DE

And finally, EH is equal to the rectangle contained by lines A and EC

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

$$\text{■} BH = \text{■} BK + \text{■} DL + \text{■} EH$$

$$\text{■} BH = BG \cdot BC, \quad \therefore \text{■} BH = A \cdot BC$$

$$\text{■} BK = BG \cdot BD, \quad \therefore \text{■} BK = A \cdot BD$$

$$\text{■} DL = DK \cdot DE, \quad \therefore \text{■} DL = A \cdot DE$$

$$\text{■} EH = EL \cdot EC, \quad \therefore \text{■} EH = A \cdot EC$$

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot BC = A \cdot BD + A \cdot DE + A \cdot EC$$

Proof:

Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

The rectangle BH is the sum of the rectangles BK DL and EH

Since BG is equal in length to A, the rectangle BH is equal to the rectangle contained by lines A and BC

Similarly, the rectangle BK is equal to the rectangle contained by lines A and BD

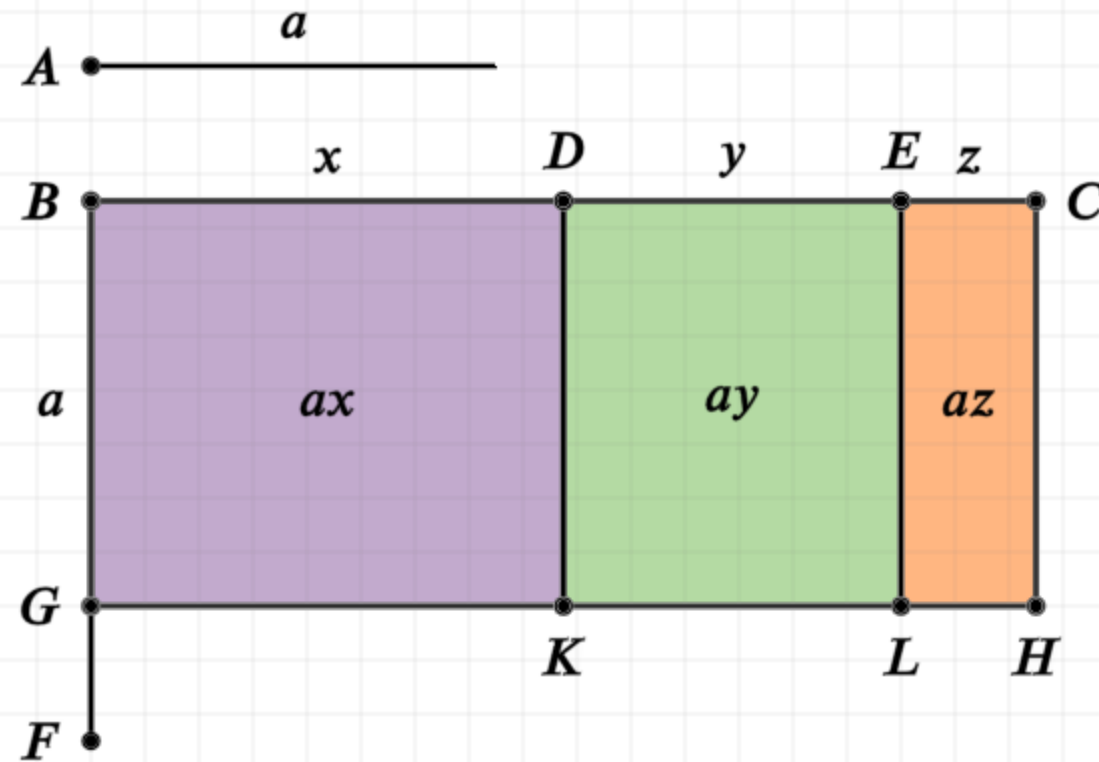
Since BG equals DK (I.34), DL is equal to the rectangle contained by lines A and DE

And finally, EH is equal to the rectangle contained by lines A and EC

Thus the rectangle formed by A,BC is equal to the sum of the rectangles formed by A,BD, A,DE and A,EC

Proposition 1 of Book 2

If there be two straight lines, and of of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles contained by the uncut straight line and each of the segments.



$$BC = BD + DE + CE$$

$$A = BG = DK = EL = CH$$

$$\text{BH} = \text{BK} + \text{DL} + \text{EH}$$

$$\text{BH} = \text{BG} \cdot \text{BC}, \quad \therefore \text{BH} = A \cdot \text{BC}$$

$$\text{BK} = \text{BG} \cdot \text{BD}, \quad \therefore \text{BK} = A \cdot \text{BD}$$

$$\text{DL} = \text{DK} \cdot \text{DE}, \quad \therefore \text{DL} = A \cdot \text{DE}$$

$$\text{EH} = \text{EL} \cdot \text{EC}, \quad \therefore \text{EH} = A \cdot \text{EC}$$

$$A \cdot \text{BC} = A \cdot \text{BD} + A \cdot \text{DE} + A \cdot \text{EC}$$

In other words

Let A and BC be two straight lines

Let BC be arbitrarily cut at points D and E

$$A \cdot \text{BC} = A \cdot \text{BD} + A \cdot \text{DE} + A \cdot \text{EC}$$

Proof:

Draw a line BF perpendicular to BC (I.11)

Define point G such that BG equals A (I.3)

Draw GH parallel to BC, and DK, EL, and CH parallel to BG (I.31)

The rectangle BH is the sum of the rectangles BK DL and EH

Since BG is equal in length to A, the rectangle BH is equal to the rectangle contained by lines A and BC

Similarly, the rectangle BK is equal to the rectangle contained by lines A and BD

Since BG equals DK (I.34), DL is equal to the rectangle contained by lines A and DE

And finally, EH is equal to the rectangle contained by lines A and EC

Thus the rectangle formed by A,BC is equal to the sum of the rectangles formed by A,BD, A,DE and A,EC

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